Session 2025-26 (ODD)

Teacher Assessment Examination (TAE)-I: Project Based Learning

Course Name & Code: MDM-III: Embedded for IoT (N-MDMIT501T)

Year/Semester: III/V

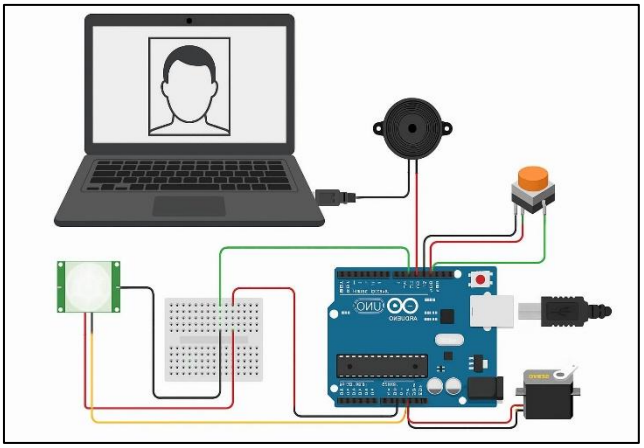
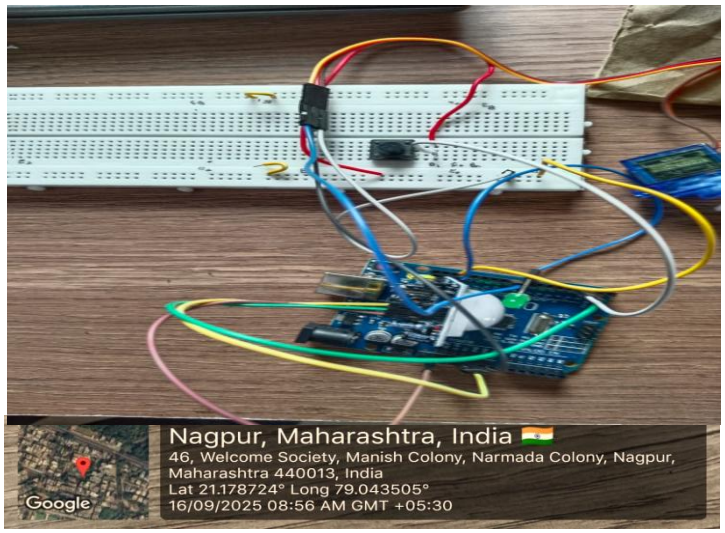
Name of Students: Riddhi Upadhyaya		
USN: CS23088	**Class Roll No. 21**	
Problem Statement: Smart Door Access Control System using Face Recognition and PIR Sensor	**Due Date of Submission**	**Actual Date of Submission**
Stage-I: Problem Statement	15/09/2025	
Stage-II: Design and Implementation	06/10/2025	

Evaluation Stages	**Stage-I**	**Stage-II**	**Total Marks**
Marks Obtained			
Max. Marks	**02**	**03**	**05**

Name & Signature of Course Incharge
(Dr. Parag Puranik)

PBL Report

Problem Statement: Design a hardware model using Arduino/ NodeMCU/Raspberry Pi.	
Stage-I: Problem Statement	
Introduction	The project focuses on developing a Smart Door Access Control System that ensures home security using Face Recognition and a PIR sensor. The system identifies authorized individuals using computer vision and unlocks the door automatically through a servo motor. If an unknown person is detected, the system captures their image and sends an email alert to the owner. This combination of IoT, image processing, and human detection provides an intelligent and secure solution to unauthorized access problems.
List of Components Used	• Arduino Uno (microcontroller for controlling PIR, LED, and Servo) • Web Camera / Laptop Camera (for face recognition) • PIR Sensor (to detect human presence/motion) • Servo Motor (for automatic door opening/closing) • LED (status indicator for door operation) • Jumper Wires & Breadboard (connections) • Power Supply / USB Cable • Laptop/PC with Python (OpenCV + Email service)
Implementation Phases/ Flowchart	**Implementation Steps:** • Capture face using camera. • Compare with known dataset using DeepFace. • If face is recognized → send "OPEN" command to Arduino. • Arduino activates Servo Motor → door opens. • If face is unknown → Arduino checks PIR status. • If PIR detects no one inside, capture face image → send email alert with photo. • System resets and waits for the next person.
Stage-II: Design & Implementation	

Interfacing Diagram (Simulation Diagram)	
Photograph of Hardware Model Developed	

Code/Program	**Python code:**
	import cv2
	from deepface import DeepFace
	import os
	import serial
	import time
	import smtplib
	from email.mime.text import MIMEText
	from email.mime.multipart import MIMEMultipart
	from email.mime.base import MIMEBase
	from email import encoders
	
	# ===== Arduino Serial Connection =====
	arduino_port = "COM6" # Replace with your COM port
	baud_rate = 9600

```
arduino = None
while arduino is None:
    try:
        arduino = serial.Serial(arduino_port, baud_rate, timeout=1)
        time.sleep(2)
        print("Arduino connected")
    except Exception as e:
        print("Waiting for Arduino...", e)
        time.sleep(2)

# ===== Email Setup =====
EMAIL_ADDRESS = "riddhiupadhyaya2005@gmail.com"
EMAIL_PASSWORD = "nmet pduq wdpj isga"  # App password
TO_EMAIL = "riddhiupadhyaya2005@gmail.com"

def send_email(subject, body, image_path=None):
    msg = MIMEMultipart()
    msg["From"] = EMAIL_ADDRESS
    msg["To"] = TO_EMAIL
    msg["Subject"] = subject

    # Add text body
    msg.attach(MIMEText(body, "plain"))

    # Attach image file
    if image_path and os.path.exists(image_path):
        with open(image_path, "rb") as f:
            part = MIMEBase("application", "octet-stream")
            part.set_payload(f.read())
        encoders.encode_base64(part)
        part.add_header(
            "Content-Disposition",
            f"attachment; filename="{os.path.basename(image_path)}"
        )
        msg.attach(part)

    with smtplib.SMTP_SSL("smtp.gmail.com", 465) as server:
        server.login(EMAIL_ADDRESS, EMAIL_PASSWORD)
        server.sendmail(EMAIL_ADDRESS, TO_EMAIL, msg.as_string())

# ===== Load Known Faces =====
known_faces_folder = "known_people"
known_faces = []
known_names = []

for filename in os.listdir(known_faces_folder):
    if filename.lower().endswith((".jpg", ".jpeg", ".png")):
```

```
        img_path = os.path.join(known_faces_folder, filename)
        known_faces.append(img_path)
        known_names.append(os.path.splitext(filename)[0])

# ===== Camera Setup =====
cap = cv2.VideoCapture(0)
last_identity = None
stable_count = 0
STABILITY_THRESHOLD = 5
door_triggered = False

while True:
    ret, frame = cap.read()
    if not ret:
        break

    try:
        detections = DeepFace.extract_faces(frame, enforce_detection=False)
        if detections:
            for det in detections:
                facial_area = det['facial_area']
                x, y, w, h = facial_area['x'], facial_area['y'], facial_area['w'],
facial_area['h']
                face_img = frame[y:y + h, x:x + w]

                identity = "Unknown"
                for idx, known_face_path in enumerate(known_faces):
                    try:
                        result = DeepFace.verify(face_img, known_face_path,
enforce_detection=False)
                        if result["verified"]:
                            identity = known_names[idx]
                            break
                    except:
                        continue

                # Stability check
                if identity == last_identity:
                    stable_count += 1
                else:
                    stable_count = 0
                last_identity = identity

                # Trigger Arduino only when stable
                if stable_count >= STABILITY_THRESHOLD:
                    if identity != "Unknown" and not door_triggered:
                        print(f"[INFO] Known person detected: {identity}")
                        try:
```

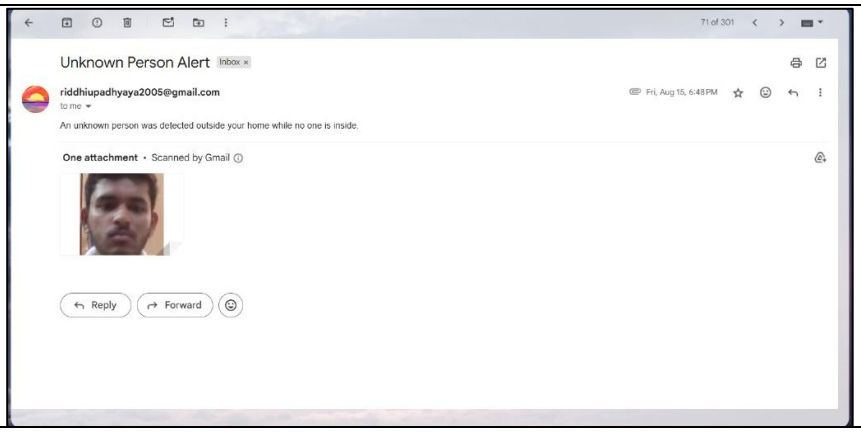
```
                    arduino.write(b'OPEN\n')
                    time.sleep(0.2)
                    door_triggered = True  # prevent repeated command
                except Exception as e:
                    print("Error sending OPEN command:", e)
            elif identity == "Unknown":
                door_triggered = False  # reset for next detection
                try:
                    arduino.write(b'CHECK\n')
                    time.sleep(0.2)
                    pir_status = arduino.readline().decode().strip()
                    print(f"[INFO] PIR status: {pir_status}")

                    if pir_status == "NO_MOTION":
                        # Capture unknown person image
                        img_name = f"unknown_{int(time.time())}.jpg"
                        cv2.imwrite(img_name, face_img)

                        # Send email alert
                        send_email(
                            "Unknown Person Alert",
                            "An unknown person was detected outside your home
while no one is inside.",
                            img_name
                        )
                        print(f"[INFO] Email sent with image: {img_name}")

                except Exception as e:
                    print("Error checking PIR / sending email:", e)

            # Draw rectangle & label
            color = (0, 255, 0) if identity != "Unknown" else (0, 0, 255)
            cv2.rectangle(frame, (x, y), (x + w, y + h), color, 2)
            cv2.putText(frame, identity, (x, y - 10),
cv2.FONT_HERSHEY_SIMPLEX, 0.9, color, 2)

    except Exception as e:
        print("Error:", e)

    cv2.imshow("Face Recognition", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
arduino.close()
```

Arduino code:

```
#include <Servo.h>

// Pin definitions
const int pirPin = 2;      // PIR sensor OUT → D2
const int ledPin = 8;      // LED (+) → D8
const int servoPin = 9;    // Servo signal → D9

Servo doorServo;
bool doorOpen = false;       // Track servo state
bool motionDetected = false; // PIR status

void setup() {
  Serial.begin(9600);
  pinMode(pirPin, INPUT);
  pinMode(ledPin, OUTPUT);

  doorServo.attach(servoPin);
  doorServo.write(0); // Door starts closed

  Serial.println("Arduino Ready");
}

void loop() {
  // Continuously monitor PIR
  motionDetected = false;
  for (int i = 0; i < 5; i++) {   // Read PIR multiple times quickly
    if (digitalRead(pirPin) == HIGH) {
      motionDetected = true;
      break;
    }
    delay(50);
  }

  // Handle serial commands from Python
  if (Serial.available() > 0) {
    String command = Serial.readStringUntil('\n');
    command.trim();
    Serial.print("Received: "); Serial.println(command);

    if (command == "OPEN" && !doorOpen) {
      openDoor();
      digitalWrite(ledPin, HIGH);  // LED ON for known person
      delay(500);
      digitalWrite(ledPin, LOW);
    }
    else if (command == "CHECK") {
```

```
      // Immediately respond with current PIR status
      Serial.println(motionDetected ? "MOTION" : "NO_MOTION");
    }
  }
}

void openDoor() {
  doorOpen = true;
  Serial.println("Opening door...");
  doorServo.write(90);     // Open door
  delay(3000);             // Keep door open
  doorServo.write(0);      // Close door
  doorOpen = false;
  Serial.println("Door closed");
}
```

Output



Conclusion

The project successfully demonstrates a Smart Door Access Control System combining face recognition and PIR-based motion sensing. The system ensures that only authorized individuals can unlock the door, while unknown entries are logged and reported via email. This enhances home security and provides a reliable, automated solution for access control.

Remarks (if any):

Name & Signature of Course Incharge
(Dr. Parag Puranik)